

GANDAKI UNIVERSITY

Faculty of Science and Technology
Bachelor of Computer Application (BCA)

WEB TECHNOLOGY

CHAPTER 2

Basic Concepts of the Web

Complete Theory, Examples & Exam Q&A

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1.1 Introduction to the Internet & WWW

The Internet and the World Wide Web are two foundational technologies that power modern digital communication. While the terms are often used interchangeably, they refer to different things.

1.1.1 What is the Internet?

The Internet is a global network of millions of interconnected computers and devices that communicate with each other using a standardized set of rules called protocols, primarily TCP/IP (Transmission Control Protocol / Internet Protocol).

The Internet was originally developed in the 1960s as ARPANET, a project of the United States Department of Defense, to enable secure communication between research institutions. Over time, it evolved into the public infrastructure we use today.

- It is a physical/logical infrastructure — wires, cables, routers, switches, and wireless links.
- It connects billions of devices worldwide, including computers, smartphones, servers, and IoT devices.
- It supports multiple services: Web (HTTP/HTTPS), email (SMTP/POP/IMAP), file transfer (FTP), video calling, and more.
- Internet Service Providers (ISPs) provide access to the internet.

NOTE: The Internet is the physical and logical network infrastructure.
The WWW, Email, FTP, VoIP are all SERVICES that run on top of the Internet.

1.1.2 What is the World Wide Web (WWW)?

The World Wide Web (WWW), invented by Tim Berners-Lee in 1989, is a system of interlinked hypertext documents and multimedia content that can be accessed via the Internet using a web browser and the HTTP protocol.

- The Web is a service, not the Internet itself.
- Web pages are written in HTML and accessed using a URL (Uniform Resource Locator).
- Web pages can contain text, images, videos, links, forms, and interactive elements.
- The Web operates using the client-server model: the browser (client) requests pages from a web server.

1.1.3 Web Browsers

A web browser is software that retrieves, interprets, and displays web pages and other resources from the Internet. It acts as the client in the client-server model.

Popular Web Browsers	Browser Functions
Google Chrome	Send HTTP requests to web servers
Mozilla Firefox	Receive and render HTML/CSS/JS

Microsoft Edge	Display text, images, multimedia
Apple Safari	Manage bookmarks and history
Opera	Support browser extensions/plugins
Brave	Handle cookies and sessions

1.1.4 Web Servers

A web server is a computer system that stores, processes, and delivers web pages to clients (browsers) over the Internet using the HTTP or HTTPS protocol. When a user enters a URL in a browser, the request is sent to the web server which responds with the requested web page.

- Popular web server software: Apache, Nginx, Microsoft IIS, LiteSpeed.
- Web servers can host static pages (HTML files) or dynamic pages generated by server-side languages like PHP, Python, Node.js, etc.
- A web server listens on port 80 (HTTP) or port 443 (HTTPS) for incoming requests.

1.1.5 URL (Uniform Resource Locator)

A URL is the complete address used to identify and access a specific resource on the internet. Every webpage, image, file, or API endpoint has a unique URL.

`https://www.example.com/page?id=1`

https:// Protocol — defines how data is transferred (HTTP = unsecured, HTTPS = SSL/TLS secured)
www Subdomain — optional prefix (www = World Wide Web subdomain)
example Second-Level Domain (SLD) — the unique name registered by the owner
.com Top-Level Domain (TLD) — indicates type/country (.com, .org, .edu, .np)
/page Path — identifies the specific resource on the server
?id=1 Query string — additional parameters passed to the server

□ **NOTE:** Internet vs. WWW Summary:

Internet = global physical network infrastructure (TCP/IP, routers, cables)
 WWW = a service running over the internet using HTTP that serves hyperlinked web pages
 Email, FTP, VoIP are also internet services but are NOT part of the WWW

1.2 Internet Protocols

A protocol is a set of standardized rules and conventions that govern how data is transmitted and received over a network. Every internet service uses one or more protocols to ensure reliable, consistent communication between systems.

1.2.1 Overview of Internet Protocols

The protocols covered in this course are HTTP, HTTPS, POP, SMTP, FTP, and WAP. Each protocol serves a specific communication function.

Protocol	Full Name	Purpose	Port
HTTP	HyperText Transfer Protocol	Transfers web pages between client and server (unsecured)	80
HTTPS	HTTP Secure	Encrypted HTTP using SSL/TLS — secured data transfer	443
POP	Post Office Protocol	Downloads emails from mail server to local client	110
SMTP	Simple Mail Transfer Protocol	Sends / relays emails between mail servers	25 / 587
FTP	File Transfer Protocol	Transfers files between computers over a network	20 / 21
WAP	Wireless Application Protocol	Provides internet access on mobile and wireless devices	—

1.2.2 HTTP — HyperText Transfer Protocol

HTTP is the foundation of data communication on the Web. It defines how clients (browsers) request resources from servers, and how servers respond.

- HTTP follows the request-response model: client sends a request, server sends back a response.
- HTTP is stateless — each request is independent and the server does not remember previous requests.
- HTTP methods: GET (retrieve data), POST (submit data), PUT (update), DELETE (remove).
- HTTP response codes: 200 OK, 301 Redirect, 404 Not Found, 500 Server Error.

1.2.3 HTTPS — HTTP Secure

HTTPS is the secure version of HTTP. It uses SSL (Secure Sockets Layer) or TLS (Transport Layer Security) to encrypt all data transmitted between the browser and the server.

- Prevents eavesdropping, man-in-the-middle attacks, and data tampering.
- Websites with HTTPS show a padlock icon in the browser address bar.
- Required for websites that handle passwords, payments, or personal data.
- Uses digital certificates issued by Certificate Authorities (CAs) to verify server identity.

1.2.4 POP — Post Office Protocol

POP (currently POP3 — version 3) is used to download emails from a remote mail server to a local client. Once downloaded, emails are typically deleted from the server.

- Suitable for users who access email from one device only.
- Works well with limited server storage.
- Disadvantage: Emails are not synchronized across multiple devices.

1.2.5 SMTP — Simple Mail Transfer Protocol

SMTP is used to send emails from a client to a mail server, or to relay emails between mail servers. It works in combination with POP or IMAP for receiving.

- SMTP handles the outgoing mail queue.
- Port 25 is used for server-to-server relay; Port 587 is used for client-to-server submission.
- POP and SMTP together form the complete email system: POP receives, SMTP sends.

1.2.6 FTP — File Transfer Protocol

FTP is a standard network protocol used to transfer files from one computer to another over a TCP/IP network. It uses two channels: a command channel (port 21) and a data channel (port 20).

- Used by web developers to upload website files to a hosting server.
- Supports authentication with a username and password.
- SFTP (Secure FTP) and FTPS add encryption for secure file transfers.

1.2.7 WAP — Wireless Application Protocol

WAP was developed in the late 1990s to enable mobile phones and wireless devices to access internet content. It allowed phones with limited processing power to view simplified web pages.

- WAP used a simplified markup language called WML (Wireless Markup Language).
- Was widely used before modern smartphones and 3G/4G networks became available.
- Largely replaced by modern mobile browsers that support full HTML5/CSS3/JS.

NOTE: Memory Tip: HTTP/HTTPS are for Web browsing. SMTP sends email. POP receives email.

FTP transfers files. WAP was for early mobile internet access.

1.3 Domain Names & Web Hosting

1.3.1 What is a Domain Name?

A domain name is a human-readable address used to identify a website or web resource on the internet. Since computers communicate using IP addresses (e.g., 142.250.190.46), domain names exist to make addresses memorable and user-friendly.

Domain Name: A human-readable label mapped to an IP address via DNS (e.g., www.google.com)

IP Address: A unique numerical label (e.g., 192.168.1.1) assigned to every device on a network

DNS: Domain Name System — the internet's distributed database that translates domain names into IP addresses

When you type www.google.com, your browser asks a DNS resolver to find the corresponding IP address, then connects to that IP to retrieve the web page.

1.3.2 Domain Name Hierarchy

Domain names follow a hierarchical structure, read from right to left. The hierarchy from highest to lowest is:

Level	Name	Description	Examples
1 (highest)	Root	Invisible "." at the top of DNS hierarchy	". " (invisible)
2	Top-Level Domain (TLD)	Broadest category; managed by ICANN	.com .org .edu .gov .np
3	Second-Level Domain	Registered unique name that identifies the site	google, facebook, gandaki
4	Subdomain	Optional prefix for organizing sections of a site	www, mail, blog, api

1.3.3 Domain Name Registration Process

To use a domain name for a website, it must be registered through an accredited registrar. The domain registration process involves the following steps:

- Step 1: Think of a unique, memorable domain name that represents your website or brand.
- Step 2: Check availability using a domain registrar's search tool (e.g., Namecheap, GoDaddy, Name.com).
- Step 3: Choose the appropriate TLD (.com for commercial, .org for organizations, .edu for education, .np for Nepal).
- Step 4: Register the domain by paying an annual fee (typically \$10–\$20/year for .com).
- Step 5: Configure DNS settings — point the domain to your web hosting server's IP address.
- Step 6: Renew annually to keep ownership; unexpired domains expire and become available to others.

1.3.4 Common Top-Level Domains (TLDs)

Generic TLDs	Country-Code TLDs
.com — Commercial	.np — Nepal
.org — Organizations	.in — India
.edu — Education	.us — United States
.gov — Government	.uk — United Kingdom
.net — Network providers	.au — Australia
.mil — Military	.cn — China
.int — International orgs	.io — (popular for tech startups)

1.3.5 Web Hosting

Web hosting is a service that provides the technology and server space needed for a website to be accessible on the internet. A web host stores your website files (HTML, CSS, images, databases) on powerful servers that are connected to the internet 24/7.

Types of Web Hosting

Shared Hosting — Multiple websites share the same server. Affordable but limited resources. Best for small/personal sites.

VPS Hosting (Virtual Private Server) — A physical server is divided into virtual servers. More control and dedicated resources than shared hosting.

Dedicated Server Hosting — Entire physical server for one client. Maximum performance, full control. Expensive — suited for large-traffic websites.

Cloud Hosting — Resources distributed across multiple servers. Highly scalable and reliable. Providers: AWS, Google Cloud, Microsoft Azure.

Managed Hosting — The hosting provider handles server management, security updates, and backups. Ideal for non-technical users.

Free Hosting — GitHub Pages, Netlify, Vercel. Great for static sites, portfolios, and learning projects with limited traffic.

1.3.6 How Domain Names and Hosting Work Together

Domain names and web hosting are two separate services that must be linked for a website to be accessible:

- You register a domain name (e.g., mysite.com) through a registrar.
- You purchase a hosting plan and get a server IP address from the hosting provider.
- You update your domain's DNS records (specifically the "A Record") to point to your server's IP address.
- When a visitor types your domain, DNS resolves it to the IP address, and the browser connects to your hosting server.
- The server responds with the website files, which the browser renders for the visitor.

NOTE: Domain name ≠ Web hosting. Domain = address. Hosting = the physical space where website files live.

You can buy a domain from one company and hosting from another — just update DNS records to link them.

Chapter 1 — Summary & Key Points

Topic 1.1 — Internet, WWW, Browsers, Servers, URL

Internet Global physical network; uses TCP/IP; supports many services (Web, Email, FTP)

WWW A service on the internet; uses HTTP; consists of interlinked web pages

Web Browser Client software that requests and renders web pages (Chrome, Firefox, Edge)

Web Server Stores and serves web pages; listens on port 80 (HTTP) or 443 (HTTPS)

URL Address of a web resource: Protocol + Subdomain + Domain + TLD + Path + Query

Topic 1.2 — Internet Protocols

HTTP (port 80) Transfers web pages; request-response model; stateless

HTTPS (port 443) Encrypted HTTP; uses SSL/TLS; required for secure websites

POP (port 110) Downloads email to local device; deletes from server

SMTP (port 587) Sends and relays email between servers

FTP (port 20/21) Transfers files between computers over network

WAP Early wireless internet for mobile devices; replaced by modern mobile browsers

Topic 1.3 — Domain Names & Web Hosting

Domain Name Human-readable address mapped to IP via DNS (e.g., www.google.com)

DNS Translates domain names to IP addresses — the internet's phone book

TLD Top-level domain: .com .org .edu .gov .np etc.

Registration Register via a registrar, choose TLD, pay fee, configure DNS

Web Hosting Server space for website files; types: Shared, VPS, Dedicated, Cloud, Free

Domain vs Hosting Domain = address; Hosting = server space; both needed for a live website

End of Chapter 1

Next: Chapter 2 — HTML and XHTML (8 hrs)

Gandaki University | Web Technology | 3 Credit Hours